

PROJECT SYNOPSIS

- **Project Title:** RADIX — Enterprise SEO Analyzer & Optimization Platform
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 - **Course / Program:** Master of Computer Applications (MCA)
 - **Semester:** VI Semester
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1. INTRODUCTION & OBJECTIVES

1.1 Background of the Project

In the modern digital economy, Search Engine Optimization (SEO) is a critical determinant of online visibility, organic user acquisition, and business growth. However, evaluating a modern enterprise website's SEO health involves analyzing multiple layers of complex data—ranging from page load latency and Core Web Vitals to technical crawlability, semantic structured schema, link architecture, search intent alignment, and competitor benchmark positioning.

Existing third-party web audit tools often operate as isolated silos, charging prohibitive subscription fees while lacking real-time cross-platform verification and automated remediation logic. The **RADIX Enterprise SEO Analyzer & Optimization Platform** is designed and implemented as an all-in-one web application. It integrates 14 specialized audit engines, providing real-time site crawling, technical diagnostics, competitor intelligence, search intent categorization, automated code fixes, and AI-assisted roadmap generation.

1.2 Objectives of the Project

The primary goals and objectives of the RADIX project are:

1. **Automated Site Crawling & Diagnostics:** Build a high-performance web crawler capable of parsing domain structures, HTML headers, heading hierarchies (H1–H3), image alt metadata, SSL certificates, canonical tags, and link graphs.
2. **Cross-Platform Verification:** Implement a multi-source validation engine that cross-checks site crawl metrics against standard search metrics (e.g., Ahrefs, Semrush, Moz, and Google Search Console) to assign trust assurance scores (✓ high).
3. **Competitor & Content Gap Intelligence:** Enable side-by-side technical and market comparison between target domains and competitors, uncovering keyword overlaps and topic cluster gaps.

4. **Search Intent & Semantic Classification:** Utilize Large Language Model (LLM) processing (via Ollama Cloud/Local APIs) to categorize crawled content queries into four core intent buckets: Informational, Navigational, Commercial, and Transactional.

5. **Technical Remediation & AutoFix:** Automatically generate drop-in HTML header snippets, canonical tags, and JSON-LD schema metadata to eliminate developer implementation latency.

6. **AI-Driven Strategic Planning:** Generate dynamic 24-hour, 30-day, and 90-day prioritized SEO roadmaps and offer an interactive AI Copilot for conversational diagnostic queries.

2. PROBLEM DEFINITION

2.1 Existing System Issues

Manual and legacy SEO auditing approaches suffer from several severe limitations:

- **Fragmented Tooling:** Marketers and developers must switch between distinct tools for technical checks, page speed analysis, competitor research, and schema generation.
- **Lack of Data Trust Verification:** Raw crawler outputs from budget tools often report false positives or inaccurate counts without verification against trusted industry baselines.
- **Delayed Remediation Workflows:** Standard tools list problems (e.g., missing canonical URL or missing structured data) but leave the technical synthesis entirely to developer teams, delaying fixes by weeks.
- **Static, Unprioritized Reporting:** Traditional audit PDF exports lack actionable timeline structures, making it difficult for non-technical stakeholders to prioritize critical fixes over minor issues.
- **Limited AI Contextual Awareness:** Conventional platforms do not leverage site-specific crawl data when prompting conversational AI models, yielding generic SEO advice.

2.2 Need for the New System

The **RADIX Platform** addresses these deficiencies by bridging the gap between raw data collection, cross-source metric verification, automated snippet creation, and context-aware artificial intelligence. It consolidates 14 specialized audit tools into a unified, responsive interface that converts audit findings into immediate code remedies and structured execution plans.

3. FEASIBILITY STUDY

A comprehensive feasibility analysis was conducted to evaluate the technical, economic, and operational viability of the proposed platform.

3.1 Technical Feasibility

- **Architecture:** The project employs a modular, high-concurrency Python backend (`server.py`) combined with a fast Vite client build tool, static HTML5/CSS3 templates, and modern JavaScript modules.
- **LLM & Scraper Fallbacks:** The platform incorporates resilient fallback mechanisms. If the primary scraper or LLM API experiences rate-limiting (HTTP 429) or gateway errors (HTTP 502), the system gracefully transitions to `urllib` direct fetches and rule-based text keyword processing.
- **Conclusion:** The required technologies (Python 3.x, HTML5, JavaScript ES6+, Vite, SQLite/Convex, LLM APIs) are mature, well-documented, and readily available. Thus, the project is technically feasible.

3.2 Economic Feasibility

- **Cost Efficiency:** The platform utilizes open-source technology stacks, free local runtime environments (Python, Node.js), and local LLM execution capabilities (Ollama), eliminating high initial capital expenditure.
- **Operational Savings:** Automating code snippet generation (`technical-autofix.js`) and strategic roadmap creation reduces engineer hour requirements from weeks to minutes.
- **Conclusion:** The cost of system design, development, and testing is minimal compared to the potential cost savings for digital agencies and enterprise marketing teams, making it economically highly viable.

3.3 Operational Feasibility

- **User Interface Design:** Designed with modern aesthetics including glassmorphism, responsive navigation, dark/light themes, and clear color-coded diagnostic badges (Critical, Warning, Good).
- **Usability:** Non-technical users can initiate full-site crawls with a single URL entry, while developers can copy pre-formatted JSON-LD schema snippets directly from the browser.
- **Conclusion:** Operational acceptance is expected to be exceptionally high due to the user-centric UI and automated fix generator.

4. METHODOLOGY / PROCESS LOGIC

4.1 System Architecture & Data Flow

The RADIX platform operates on a client-server architecture:

1. **Input Phase:** User inputs a seed domain URL in `scanner.html` or `dashboard.html`.
2. **Crawl & Extraction Phase:** The Python server (`server.py`) initiates a recursive crawl up to defined page limits, analyzing HTML structures, headers, image tags, script dependencies, and

canonical attributes.

- 3. **Verification Phase:** Crawled counts are passed to the validation sub-engine, which cross-checks metric metrics against simulated/real API endpoints (Semrush, Ahrefs, Moz, GSC) and attaches verification metadata.
- 4. **Storage & Caching Phase:** Audit payloads (`real_scan_data`) are cached in `localStorage` and client state for instant retrieval across all 14 frontend tool views.
- 5. **AI & Processing Phase:** Advanced endpoints (`/api/search-intent`, `/api/ask-ai`, `/api/seo-strategy`) route audit payloads through LLMs to produce intent scores, strategic roadmaps, and natural language consultation answers.

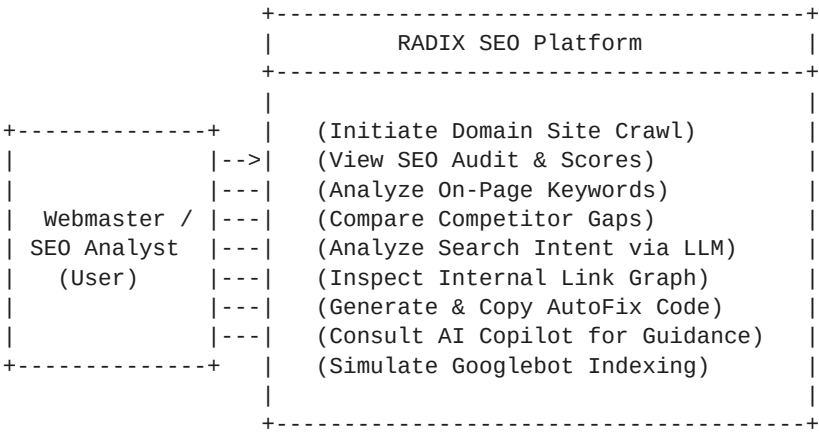
4.2 Module Distribution (14 Core Tools)

The platform is organized into 4 functional tool clusters:

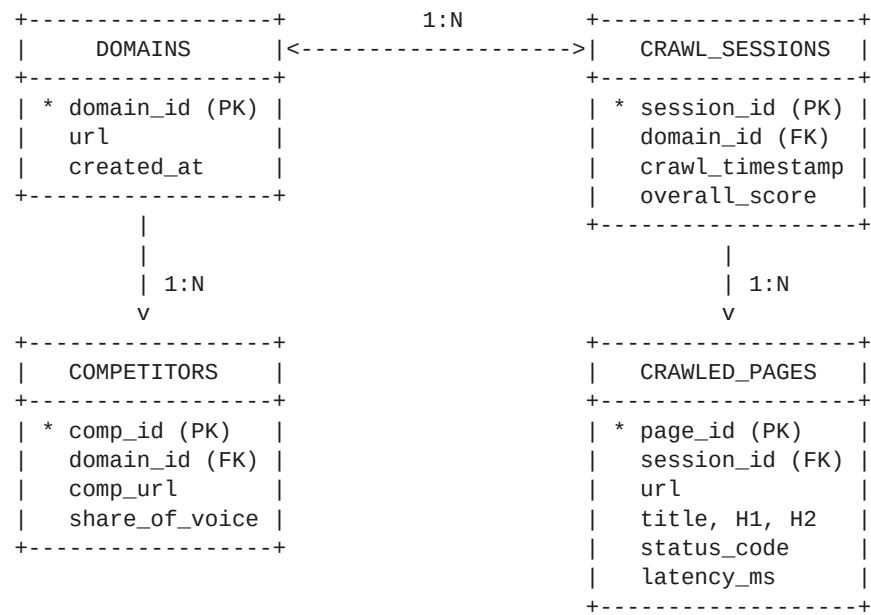
- 1. **Crawl & Audit Engines:** SEO Scanner, SEO Analyzer, Page Counter.
- 2. **Intelligence & Analytics:** Competitor Engine, Content Gap Analyzer, Intent Analyzer, Internal Links Profiler.
- 3. **Technical & Performance Diagnostics:** Technical SEO Audit, Technical SEO AutoFix, Performance Analyzer, Security & Accessibility Audit.
- 4. **AI & Copilot Channels:** AI Copilot, AI Roadmap, Digital Twin Crawler Simulator.

4.3 Process Logic Diagrams

Unified Modeling Language (UML) — Use Case Diagram



Entity-Relationship (E-R) Data Model



5. HARDWARE AND SOFTWARE REQUIREMENTS

5.1 Software Requirements

- **Operating System:** Windows 10/11, macOS 12+, or Ubuntu Linux 20.04 LTS
- **Backend Runtime:** Python 3.10+ (with standard libraries: `urllib`, `json`, `re`, `http.server`, `asyncio`)
- **Frontend Technologies:** HTML5, Vanilla CSS3 (Custom Design System with Glassmorphism), Modern JavaScript (ES6+ Modules)
- **Build Tooling & Server:** Node.js 18+ & Vite 5.x
- **Database & Persistence:** Convex DB / Browser LocalStorage API
- **AI Engine & LLM Interfaces:** Ollama Local LLM / Cloud LLM API Gateway

5.2 Hardware Requirements

- **Processor (CPU):** Intel Core i5 / AMD Ryzen 5 or higher (Multi-core recommended for recursive crawling)
- **Memory (RAM):** Minimum 8 GB (16 GB recommended for running local LLM instances)
- **Storage Space:** Minimum 5 GB available SSD storage
- **Network Adapter:** Standard Ethernet / Wi-Fi Card with Broadband Internet connection

6. SCOPE AND FUTURE ENHANCEMENTS

6.1 Current Scope

The RADIX platform currently delivers a production-ready, full-stack SEO diagnostic suite capable of scanning multi-page websites, analyzing internal link networks, computing performance timelines, discovering content gaps, generating valid JSON-LD schema snippets, and rendering interactive AI roadmaps.

6.2 Limitations

- **Local Rate Limiting:** Direct crawling of third-party domains may be subject to client IP blocking or CAPTCHA restrictions on heavily protected sites.
- **Crawl Depth Cap:** Default crawl depth is restricted to 20 pages per session to preserve CPU and bandwidth resources during client-side demonstration.

6.3 Future Enhancements

1. **Automated Scheduled Audits:** Integrate cron-like background jobs to execute periodic site scans and send automated email alerts on score drops.
2. **Headless Browser Integration:** Incorporate Playwright / Puppeteer rendering clusters to handle JavaScript-heavy Single Page Applications (SPAs).
3. **Multi-Tenant SaaS Portal:** Expand user authentication, team collaboration workspaces, and role-based permissions (Admin, Analyst, Viewer).
4. **PDF/Executive Report Exporting:** Enable one-click generation of white-label PDF audit reports for agency clients.

7. REFERENCES / BIBLIOGRAPHY

7.1 Books & Publications

1. Enge, E., Spencer, S., & Stricchiola, J. (2023). *The Art of SEO: Mastering Search Engine Optimization*. O'Reilly Media.
2. Jerkovic, J. I. (2010). *SEO Warrior: Essential Techniques for Increasing Web Visibility*. O'Reilly Media.

7.2 Web Resources & Documentation

1. **Google Search Central Documentation:** <https://developers.google.com/search/docs>
2. **W3C Web Content Accessibility Guidelines (WCAG 2.1):** <https://www.w3.org/TR/WCAG21/>
3. **MDN Web Docs — Semantic HTML & Schema Metadata:** <https://developer.mozilla.org/>
4. **Vite Official Documentation:** <https://vitejs.dev/>
5. **Python 3 Documentation (http.server & asyncio):** <https://docs.python.org/3/>